DHEERAJ

Software Developer

⌖ Panipat, Haryana | ☎ +91-8930601580 | ✉ dheerajsaroha2892002@gmail.com
in linkedin.com/in/dheeraj-saroha | </> github.com/dheerajsaroha

PROFESSIONAL SUMMARY

Software Developer with a B.Tech in Computer Science (AI & ML) and about 1.5 years of combined software engineering experience (Software Engineer, Unthinkable Solution LLP; Software Engineer Intern, NIELIT Kurukshetra). Designs, evaluates and deploys machine learning models in Python using Pandas, NumPy, Scikit-learn and XGBoost. Experienced in exploratory data analysis (EDA), data preprocessing, feature engineering, handling class imbalance (SMOTE), cross-validation and hyperparameter tuning, and in serving models through Flask REST APIs backed by PostgreSQL. Familiar with Agile/Scrum delivery, Git/GitHub and Docker.

TECHNICAL SKILLS

Languages: Python, SQL, Object-Oriented Programming (OOP), Data Structures

Machine Learning: Scikit-learn, XGBoost, PyTorch, Regression, Classification, Clustering, Feature Engineering, SMOTE, Cross-Validation, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Model Evaluation (F1, ROC AUC, Precision, Recall)

Data Processing & Visualization: Pandas, NumPy, PySpark, Data Cleaning, Data Transformation, Data Validation, Matplotlib, Seaborn

Backend, APIs & Databases: Flask, FastAPI, Ruby on Rails (RoR), WordPress, REST APIs, JSON, PostgreSQL, SQL Query Optimization, Postman

NLP & Generative AI: Natural Language Processing (NLP), Text Processing, Recurrent Neural Networks (RNN), Deep Learning, LangChain, LangGraph, Model Context Protocol (MCP), Agentic AI, Prompt Engineering

Tools & Practices: Git, GitHub, Docker, Linux, Jupyter Notebook, VS Code, PyCharm, Agile/Scrum, Debugging

ACADEMIC PROJECTS

Customer Churn Prediction System *Python, Scikit-learn, XGBoost, SMOTE, Flask, Pandas*

- Engineered an end-to-end machine learning pipeline on the Telco Customer Churn dataset, covering EDA, data preprocessing and class-imbalance analysis (~27% churn rate).
- Benchmarked 10 classification models using stratified 5-fold cross-validation and refined hyperparameters with GridSearchCV and RandomizedSearchCV.
- Applied SMOTE oversampling to lift XGBoost F1 score from 0.553 to 0.850; the final Random Forest model reached 77.4% accuracy and 82.7% ROC AUC.
- Deployed the Random Forest model as a Flask web application exposing 2 interfaces, a prediction UI and a REST API. Live demo: customer-churn-prediction-topaz.vercel.app

Healthcare Readmission Risk Prediction System *Python, PySpark, Scikit-learn, Pandas, NumPy, Flask*

- Created an end-to-end predictive analytics application on structured healthcare data, using PySpark for data processing and transformation and Pandas/NumPy for cleaning, preprocessing and feature engineering.
- Trained and evaluated a Scikit-learn classification model against standard metrics and exposed predictions through a Flask REST API. Live demo: healthcare-readmission-risk-predict-phi.vercel.app

RNN-Based Question Answer System *Python, NLP, RNN, Deep Learning*

- Designed a deep learning question-answering system using Recurrent Neural Network (RNN) architectures for sequence modeling, with NLP text-processing for input preparation and response generation. GitHub: github.com/dheerajsaroha/RNN-based-Question-Answer-System

PROFESSIONAL EXPERIENCE

Software Engineer Intern
NIELIT Kurukshetra **Feb 2026 – Aug 2026**

- Developed Python backend modules with structured data validation for an enterprise attendance management platform used by 100+ active workforce members.
- Optimized SQL queries and PostgreSQL relational schemas to speed up data retrieval while maintaining transaction reliability and consistency.
- Worked in an Agile/Scrum team over a 6-month internship, clarifying requirements, troubleshooting issues and delivering sprint tasks on schedule.

Software Engineer
Unthinkable Solution LLP **Aug 2022 – Aug 2023**

- Delivered and maintained modular web application components in Ruby on Rails (RoR) and WordPress, integrating backend services in a live production environment for 12 months.
- Partnered with cross-functional teams to gather requirements, debug issues and ship reliable software releases.

EDUCATION

B.Tech, Computer Science (AI & ML), Guru Jambheshwar University of Science and Technology | CGPA: 7.08 *2023 – 2026*
Diploma, Computer Science Engineering, Government Polytechnic Sonipat | Score: 72% *2020 – 2022*

CERTIFICATIONS

- Accenture Developer and Technology Virtual Experience Program (Forage)
- Python for Data Science and Software Engineering
- Professional Communication and Team Collaboration, NIIT